

2. Healthcare Analytics Dataset

Question 2

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

Patient ID	Age	Sugar Level	BP	BMI	Disease Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	125	26	No
P07	52	165	140	30	Yes
P08	35	105	120	24	No
P09	48	155	135	29	Yes
P10	30	95	115	23	No
P11	62	210	155	35	Yes
P12	38	115	122	25	No
P13	50	170	142	30	Yes
P14	27	88	108	21	No
P15	58	190	148	33	Yes
P16	42	130	128	27	No
P17	46	150	132	28	Yes
P18	34	100	116	24	No
P19	53	175	144	31	Yes
P20	29	92	112	22	No

Answer:

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# =====
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# Healthcare Analytics EDA (4 Charts)
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import pandas as pd
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```
import matplotlib.pyplot as plt
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# Load Dataset
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```
df = pd.read_csv("healthcare_dataset.csv")
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# -----
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# Basic Information
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# -----
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```
print("Dataset Shape:", df.shape)
```

```
print("\nMissing Values")
```

```
print(df.isnull().sum())
```

```
print("\nStatistical Summary")
```

```
print(df.describe())
```

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# =====
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# Chart 1 : Disease Status Count
# =====
plt.figure(figsize=(6,5))
df['Disease_Status'].value_counts().plot(kind='bar', color=['skyblue','salmon'])

plt.title("Disease Status Distribution", fontsize=14)
plt.xlabel("Disease Status")
plt.ylabel("Number of Patients")

plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.show()

# =====
# Chart 2 : Average Health Parameters
# =====
avg = df.groupby('Disease_Status')[['Sugar_Level','BP','BMI']].mean()

avg.plot(kind='bar', figsize=(8,5))

plt.title("Average Health Parameters by Disease Status", fontsize=14)
plt.xlabel("Disease Status")
plt.ylabel("Average Value")

plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.legend(title="Parameters")
plt.show()

# =====
# Chart 3 : Age vs Sugar Level
# =====
plt.figure(figsize=(7,5))

colors = df['Disease_Status'].map({'Yes':'red','No':'green'})

plt.scatter(df['Age'], df['Sugar_Level'],
            c=colors,
            s=90)

plt.title("Age vs Sugar Level", fontsize=14)
plt.xlabel("Age (Years)")
plt.ylabel("Blood Sugar Level")

plt.grid(True)

from matplotlib.lines import Line2D

legend_elements = [
    Line2D([0],[0], marker='o', color='w', label='Disease = Yes',

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        markerfacecolor='red', markersize=10),
    Line2D([0],[0], marker='o', color='w', label='Disease = No',
        markerfacecolor='green', markersize=10)
]

plt.legend(handles=legend_elements)
plt.show()

# =====
# Chart 4 : Correlation Heatmap
# =====
corr = df[['Age','Sugar_Level','BP','BMI']].corr()

plt.figure(figsize=(6,5))

plt.imshow(corr, cmap='coolwarm')

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Between Health Parameters", fontsize=14)

for i in range(len(corr.columns)):
    for j in range(len(corr.columns)):
        plt.text(j, i,
            f'{corr.iloc[i,j]:.2f}',
            ha='center',
            va='center',
            color='black')

plt.tight_layout()
plt.show()

# =====
# Final Findings
# =====

print("\n===== FINAL RESULT =====")
print("1. Half of the patients are diagnosed with disease, while the remaining half are healthy.")
print("2. Patients with disease have higher average Sugar Level, Blood Pressure, and BMI.")
print("3. Older patients generally show higher blood sugar levels and are more likely to have the disease.")
print("4. Sugar Level, Blood Pressure, and BMI have a positive correlation, indicating they increase together.")
print("5. The dataset contains no missing or duplicate values and is suitable for further predictive analysis.")

```

Output :

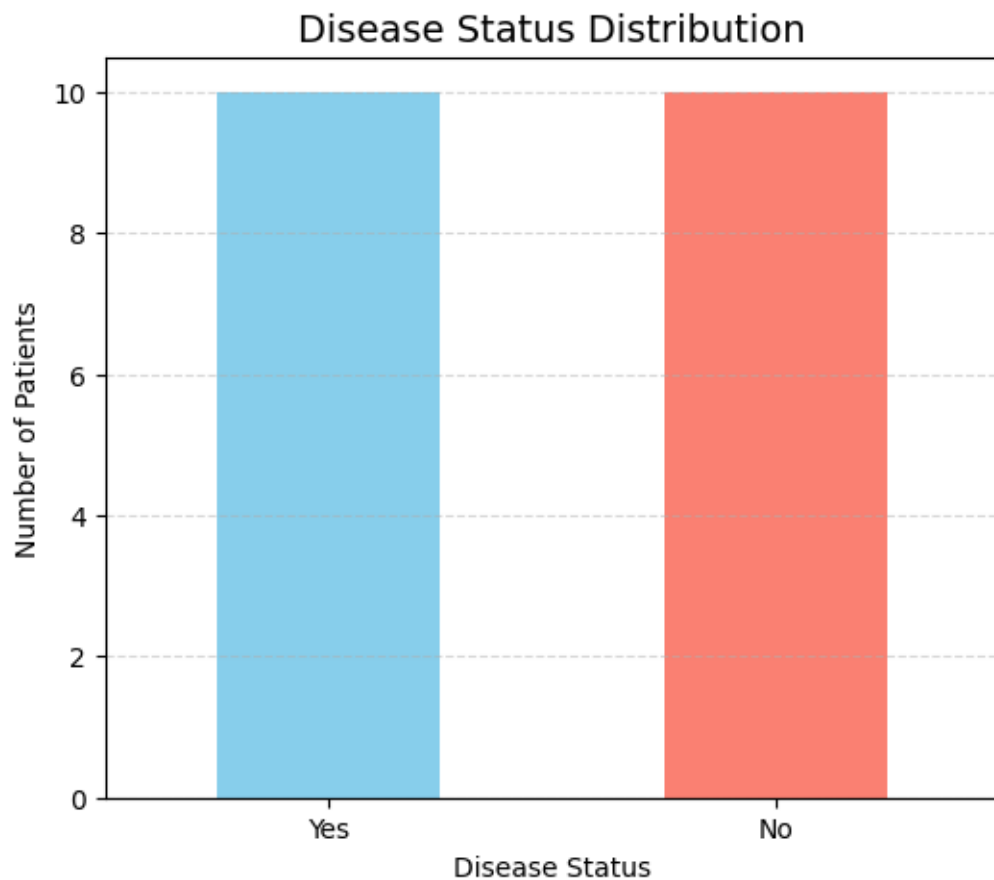
Dataset Shape: (20, 6)

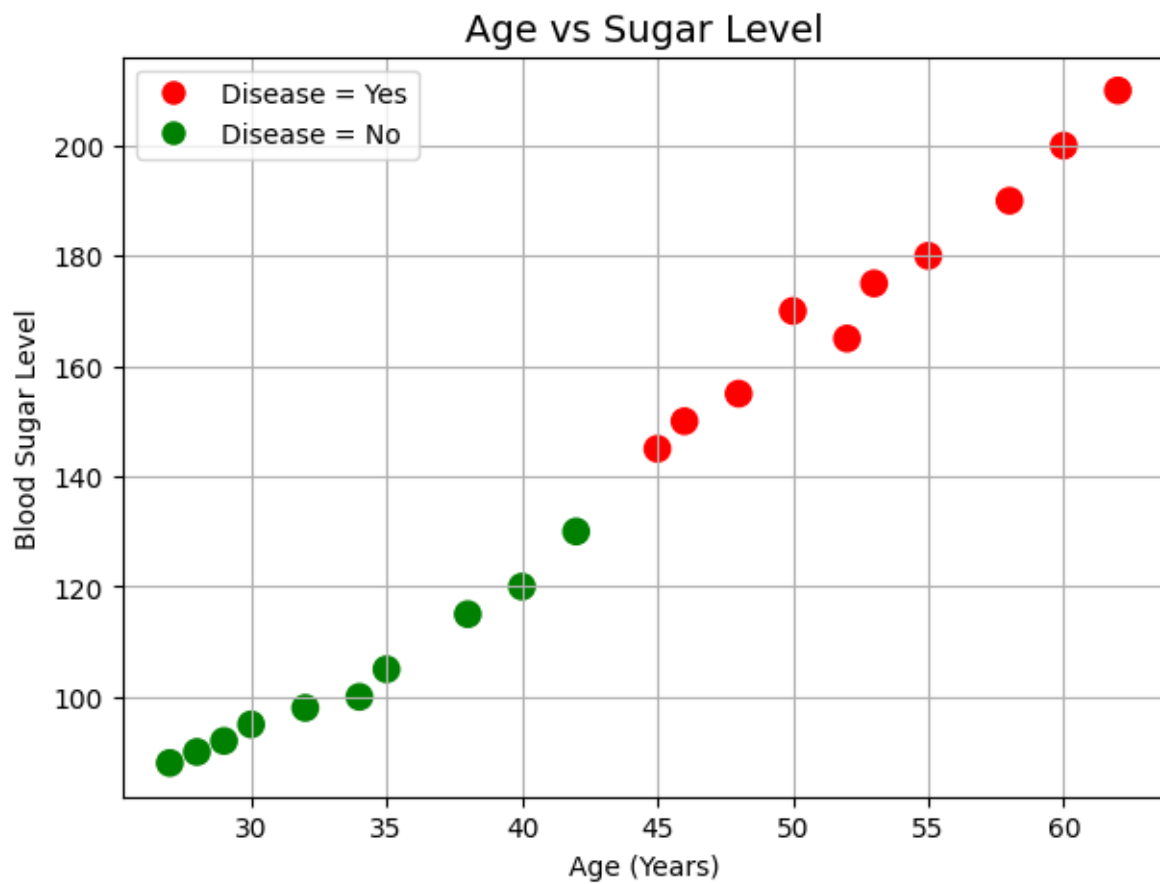
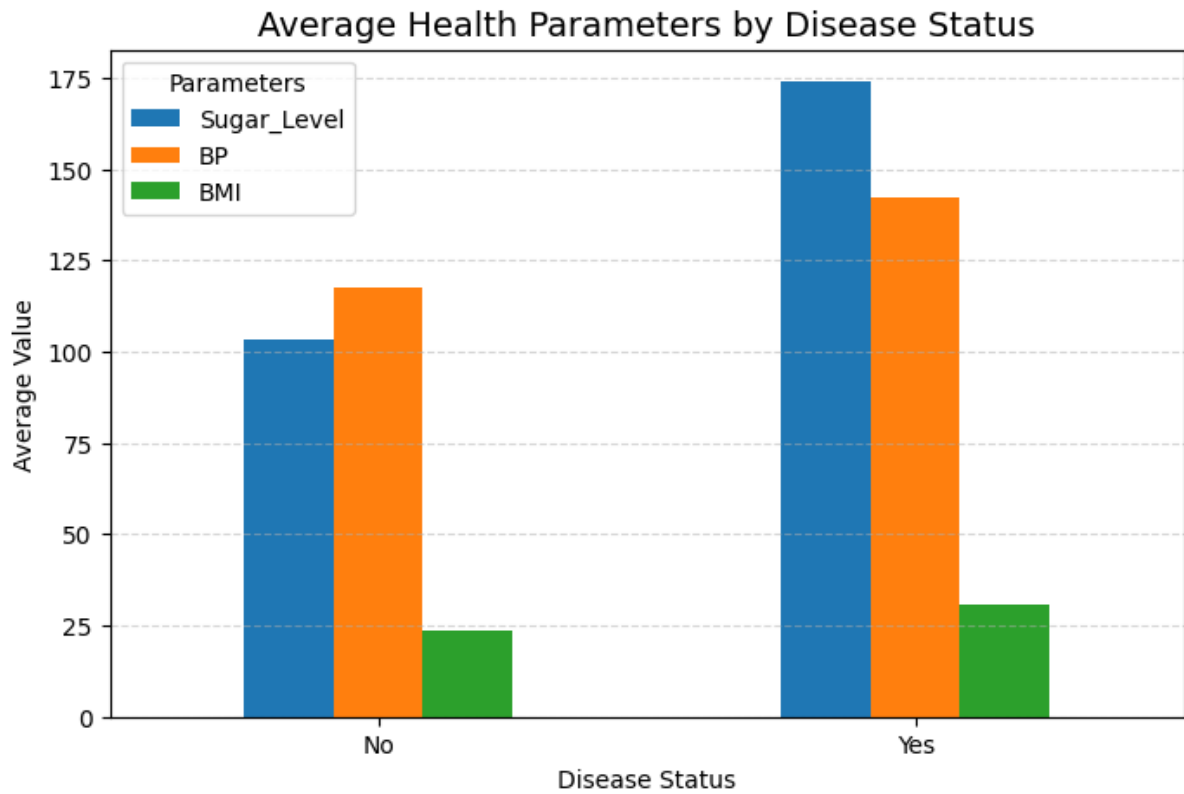
Missing Values

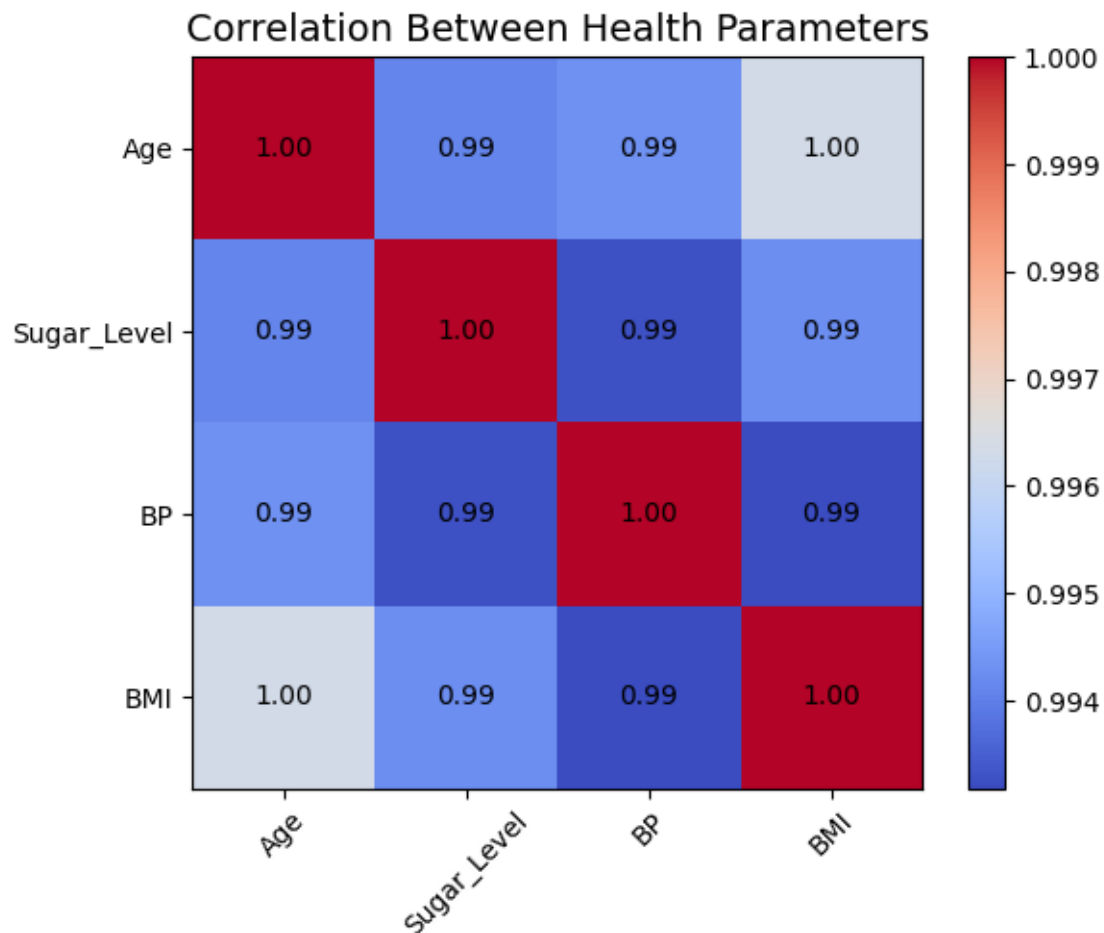
Patient_ID 0
Age 0
Sugar_Level 0
BP 0
BMI 0
Disease_Status 0
dtype: int64

Statistical Summary

	Age	Sugar_Level	BP	BMI
count	20.000000	20.000000	20.000000	20.000000
mean	43.200000	138.650000	129.750000	27.300000
std	11.316499	40.33449	14.523574	4.256265
min	27.000000	88.000000	108.000000	21.000000
25%	33.500000	99.500000	117.500000	23.750000
50%	43.500000	137.500000	129.000000	27.500000
75%	52.250000	171.250000	142.500000	30.250000
max	62.000000	210.000000	155.000000	35.000000







===== FINAL RESULT =====

1. Half of the patients are diagnosed with disease, while the remaining half are healthy.
2. Patients with disease have higher average Sugar Level, Blood Pressure, and BMI.
3. Older patients generally show higher blood sugar levels and are more likely to have the disease.
4. Sugar Level, Blood Pressure, and BMI have a positive correlation, indicating they increase together.
5. The dataset contains no missing or duplicate values and is suitable for further predictive analysis.